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TRANSFORMER OVERLOADING & LIFE SPAN

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TRANSMISSION ENGINEERING STANDARD

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1.0 SCOPE

The Scope of this standard is to highlight the different aspects and provide information regarding Overloading and its effect to Life Span of Power Transformers employed in the transmission network of Saudi Electricity Company (SEC), Kingdom of Saudi Arabia.

2.0 INTRODUCTION

This Standard is intended to be used as a guide in determining the extent of overload that oil immersed Power Transformers can carry under various operating conditions and with various degrees of insulation deterioration. Load in excess of normal rating may be required to be carried under certain conditions when necessary. For operations at loads above normal rating, on a repetitive basis, the cooling system should be maintained at maximum operating efficiency. The windings and oil in the Transformer should be checked to determine that they are absolutely clean and free from excessive amounts of moisture, sludge and internal obstructions to oil circulation such as reduced oil duct clearances, of any nature, are not present. Transformer capacity including overload capability may at times be increased by augmenting the cooling air supply with blowers or making other cooling system changes. The expected increase in the hot spot temperature shall be accounted and confirmed to be under safer limits. However, exposure towards these factors shall be useful to provide recommendations for loading the Power Transformers above the nameplate rating and guidance to choose appropriate rated quantities and loading conditions.

3.0 EFFECT OF LOADING BEYOND NORMAL RATING

A normal life expectancy is a conventional reference basis for continuous duty under design ambient temperature and rated operating conditions. The application of a load in excess of nameplate rating and/or an ambient temperature higher than design ambient temperature involves a degree of risk and accelerated ageing. These risks can be reduced by clearly specifying the maximum loading conditions that should be considered into account in the transformer design.

3.1 General Consequence of overloading

The consequences of loading a transformer beyond its nameplate rating are as follows:

a) The temperature of windings, cleats, leads, Insulation and oil will increase and can reach unacceptable levels.

b) The leakage flux density outside the core increases, causing additional eddy current heating in metallic parts linked by the leakage flux.

c) As the temperature changes, the moisture and gas content in the insulation and in the oil will change.

d) Bushings, Tap-changers, Cable-end connections and current transformers will also be exposed to higher stresses which encroach upon their design and application margins.

The combination of the main flux and increased leakage flux imposes restrictions on possible core over excitation.

As a consequence, there will be a risk of premature failure associated with the increased currents and temperatures. This risk may be of an immediate short time character that comes from the cumulative effect of thermal ageing of the insulation in the transformer over many years.

3.2 Effects and hazards of Short time emergency loading

Short time increased loading will result in a severe condition having an increased risk of failure. Short time emergency loading causes the conductor hot-spot to a level likely to result in a temporary reduction in dielectric strength. However, acceptance of this condition for a short time may be preferable to loss of supply. This type of loading is expected to occur rarely and it should be rapidly reduced or the transformer disconnected within a short time in order to avoid its failure.

The permissible duration of this load is shorter than the thermal time constant of the whole transformer and depends on the operating temperature before the increase in loading; typically, it would be less than half- an- hour.

The main risk of short time failure is the reduction in dielectric strength due to the possible presence of gas bubbles in a region of high electrical stress, that is, the windings and the leads. These bubbles are likely to occur when the hot-spot temperature exceeds 140ºC for a transformer with a winding insulation moisture content of about 2%. This critical temperature will decrease as the moisture concentration increases.

Gas bubbles can also develop (either in oil or in solid insulation) at the surface of heavy metallic parts heated by the leakage flux or be produced by the super-saturation of the oil. However, such bubbles develop in the regions of low electric stress and have to circulate in regions where the stress is higher before any significant reduction in dielectric strength occurs.

Bare metallic parts, except windings, which are not in direct thermal contact with cellulosic insulation but are in contact with non-cellulosic insulation (for example, aramid paper, glass fiber) and the oil in the transformer, may rapidly rise to higher temperatures. A temperature of 180ºC should not be exceeded.

Temporary deterioration of the mechanical properties at higher temperatures could reduce the short circuit strength.

Pressure build-up in the bushings may result in a failure due to oil leakage. Gassing in condenser type bushings may also occur if the temperature of the insulation exceeds 140ºC.

The expansion of the oil may cause overflow of the oil in the conservator.

The limitations in the maximum hot-spot temperatures in the windings, core and structure parts are based on the consideration of short term risks.

The short term risks normally disappear after the load is reduced to normal level, but they need to be clearly identified and accepted by all SEC concerned departments.

3.3 Effects of Long-time emergency loading

This is not a normal operating condition and its occurrence is expected to be rare, but it may persist for weeks or even months leading to considerable ageing.

a) Deterioration of mechanical properties of conductor insulation will accelerate at higher temperatures. If this deterioration proceeds far enough, it may reduce the effective life of the transformer, particularly if the latter is subjected to system short circuits or transportation events.

b) Other insulation parts, especially parts sustaining the axial pressure of the winding block, could also suffer increased ageing rates at higher temperature.

c) The contact resistance of the tap-changers could increase at elevated currents and temperatures and in severe cases, thermal runaway could take place.

d) The gasket materials in the transformer may become more brittle as a result of elevated temperatures.

The calculation rules of the relative ageing rates and percent loss of life are based on considerations of long-term risks.

3.4 Transformer size

The sensitivity of transformers to loading beyond name plate rating usually depends on their size. As the size increases, the tendency is that:

- The leakage flux density increases
- The short circuit forces increase
- The mass of the insulation, which is subjected to a high electric stress, is increased
- The hot-spot temperatures are more difficult to determine.

Thus, a large transformer could be more vulnerable to loading beyond name plate rating than a smaller one. In addition, the consequence of the transformer failure is more severe for larger size, than for smaller units.

Therefore, in order to apply a reasonable degree of risk for the expected duties, following three categories are considered:

a) Distribution transformers, (25kVA to 2500kVA) for which only the hot-spot temperatures in the windings and thermal deterioration shall be considered.

b) Medium power transformers, (2501kVA to 100,000kVA) where the variations in the cooling mode shall be considered.

c) Large power transformers, (above 100,000kVA) where also the effects of stray leakage flux are significant and the consequence of the failure are severe.

3.5 Non thermally and thermally upgraded insulation paper

The purpose of thermally upgraded insulation paper is to neutralize the production of acids caused by the hydrolysis (thermal degradation) of the material over the life

time of the transformer. This hydrolysis is more active in elevated temperatures and the research results referred in IEC 60076-7 indicate that thermally upgraded insulation papers retain a higher percentage of their tensile and bursting strength than the untreated papers when exposed to elevated temperatures. The same reference also show change of degree of polymerization (DP) over time of non-thermally and thermally upgraded paper exposed to temperature of 150ºC.

4.0 CALCULATION OF AGEING RATE FOR OIL IMMERSED POWER TRANSFORMER

Ageing or deterioration of insulation is a time function of temperature, moisture content, oxygen content and acid content.

The model presented in this standard is based on the insulation temperature which is considered as the controlling parameter for determination of the ageing rate for power Transformers. Since the temperature is not uniform throughout the windings, the part that is operating at the highest temperature will undergo the greatest deterioration. Therefore, the rate of ageing is referred to the winding hot spot temperature.

Relative Ageing rate "V" is determined using the following formula

$$V = 2^{(\theta_h - 98)/6}$$ for non thermally upgraded paper Eq --- (1)

$$V = e^{\left(\frac{15000}{110+273}\right) - \left(\frac{15000}{\theta_h + 273}\right)}$$ for thermally upgraded paper Eq --- (2)

Where θ_h = hot spot temperature (Refer IEC 60076-7, 2005)

Equations (1) & (2) imply that "V" is very sensitive to the hot spot temperature as can be seen from the following Table-1.

Table – 1 Relative ageing rates due to different hot spot temperature

θh ºC	Non-upgraded paper insulation V	Upgraded paper insulation V
80	0.125	0.036
86	0.25	0.073
92	0.5	0.145
98	1.0	0.282
104	2.0	0.536
110	4.0	1.0
116	8.0	1.83
122	16.0	3.29
128	32.0	5.8
134	64.0	10.1
140	128.0	17.2

5.0 LOSS OF LIFE CALCULATION FOR OIL IMMERSED POWER TRANSFORMER

The loss of life "L" over a period of time for the Oil immersed Power Transformer is determined from the following formula.

$$L = \int_{t_1}^{t_2} V dt \qquad \text{Equation ------(3)}$$

or

$$L = \sum_{n=1}^{N} V_n \times t_n \qquad \text{Equation ------(4)}$$

Where

V_n is the relative ageing rate during interval n per Equation (1) & (2).

t_n is the nth time interval; n is the number of each time interval.

N is the total number of intervals during the time considered.

5.1 Guide for temperature and current limitations

Table – 2 Current and temperature limits applicable to loading beyond name plate rating

Types of Loading	Medium Power Transformers	Large Power Transformers
Normal cyclic loading		
Current (p.u)	1.5	1.3
Winding hot-spot temperature and metallic parts in contact with cellulosic insulation material (ºC)	120	120
Other metallic hot-spot temperature (in contact with oil, aramid paper, glass fiber materials (ºC)	140	140
Top oil temperature (ºC)	105	105
Long time emergency loading		
Current (p.u)	1.5	1.3
Winding hot-spot temperature and metallic parts in contact with cellulosic insulation material (ºC)	140	140
Other metallic hot-spot temperature (in contact with oil, aramid paper, glass fiber materials (ºC)	160	160
Top oil temperature (ºC)	115	115

Types of Loading	Medium Power Transformers	Large Power Transformers
Short time emergency loading		
Current (p.u)	1.8	1.5
Winding hot-spot temperature and metallic parts in contact with cellulosic insulation material (ºC)	160	160
Other metallic hot-spot temperature (in contact with oil, aramid paper, glass fiber materials (ºC)	160	180
Top oil temperature (ºC)	115	115
Note : The temperature and current limits are not intended to be valid simultaneously. The current may be limited to a lower value than that shown in order to meet the temperature limitation requirement. Conversely, the temperature may be limited to a lower value than that shown in order to meet the current limitation requirement.		

The limits on load current, hot-spot temperature and temperature of metallic parts other than windings and leads stated in the above furnished table should not be exceeded. Further, it should be noted that when the hot-spot temperature exceeds 140ºC, gas bubbles may develop which could jeopardize the dielectric strength of the transformer.

5.2 Specific limitations for large transformers

For large transformers, additional limitations, mainly associated with the leakage flux, shall be taken into consideration.

- The combination of leakage flux and main flux in the limbs or yokes or the magnetic circuit makes large transformers more vulnerable to over excitation than smaller transformers especially when loaded above nameplate rating. Increased leakage flux may also cause additional eddy current heating of other metal parts.

- The consequences of degradation of the mechanical properties of insulation as a function of temperature and time, including wear due to thermal expansion, may be more severe for large transformers.

- Hot-spot temperatures outside the windings cannot be obtained from a normal temperature test. Even if such a test at a rated current indicates no abnormalities, it is not possible to draw any conclusions for higher currents since this extrapolation may not have been taken into account at the design stage.

- Calculation of the winding hot-spot temperature rise at higher than rated currents, based on the results of a temperature rise test at rated current, may be less reliable for large units than for smaller ones.

6.0 CALCULATION OF HOT-SPOT TEMPERATURE AT VARYING AMBIENT TEMPERATURE AND LOAD CONDITIONS

Calculation of hot spot temperature as a function of time, shall be done by the two following methods shown below:

a) Exponential equations solution, suitable for a load variation according to a step function.

This method is particularly suitable when heat transfer parameters are determined by test and it yields proper result in the following cases:

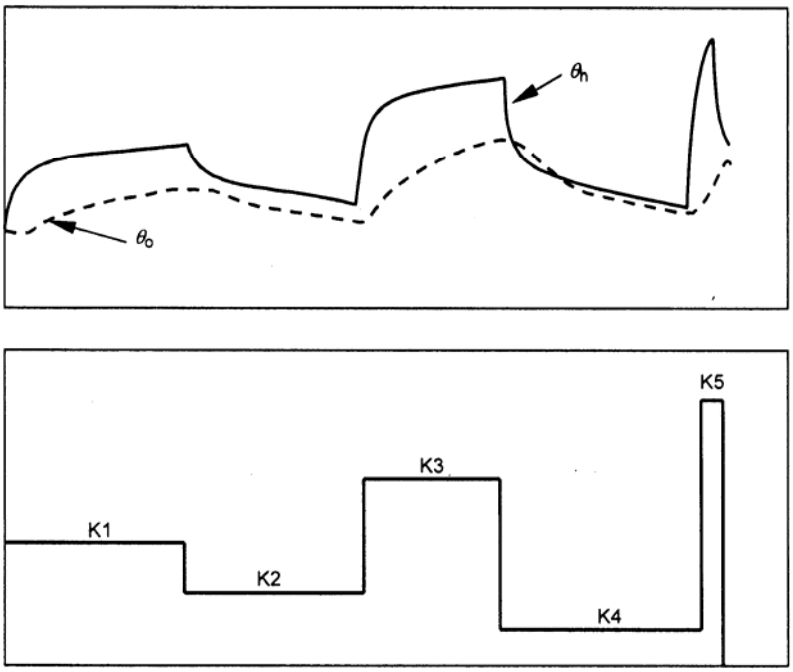
- Each of the increasing load steps is followed by a decreasing load step or vice versa.
- In case of "N" successive increasing load steps (N ≥ 2), each of the (N-1) first steps has to be long enough for the hot-spot-to-top-oil gradient "$\Delta\vartheta_h$" to obtain steady state. The same condition is valid in case of "N" successive decreasing load steps (N ≥ 2).

Note: N = Number of load steps

The practical example of the exponential equations method and its flowchart is illustrated in section 7.0.

b) Differential equations solution, suitable for arbitrarily time varying load factor "K" and time varying ambient temperature "θ_a". This method is particularly applicable for on-line monitoring, especially as it does not have any restrictions concerning the load profile.

6.1 Exponential equations solution



Key

θ_h Winding hot-spot temperature

θ_o Top oil temperature in tank

K1 is 1.0 K2 is 0.6 K3 is 1.5 K4 is 0.3 K5 is 2.1

Fig : 1 Temperature response to step changes in the load current

The hot-spot temperature is equal to the sum of the ambient temperature, the top oil temperature rise in tank and the temperature difference between the hot-spot and top-oil in the tank.

The temperature increase to a level corresponding to load factor "K" is given by equation (5):

$$\theta_h(t) = \theta_a + \Delta\theta_{oi} + \left\{ \Delta\theta_{or} \times \left[\frac{1 + R \times K^2}{1 + R} \right]^x - \Delta\theta_{oi} \right\} \times f_1(t) + \Delta\theta_{hi} + \left\{ Hg_r K^y - \Delta\theta_{hi} \right\} \times f_2(t)$$

Where, "θ_h"- Hot spot temperature, "θ_a" – Ambient temperature, "$\Delta\theta_{oi}$"- Top-oil (in tank) temperature rise at start, "$\Delta\theta_{or}$" – Top-oil (in tank) temperature rise in steady state at rated losses, "R"- Ratio of load losses at rated current to no-load losses, "K"- Load factor, "x"- Oil exponent, "y"- Winding exponent, "$\Delta\theta_{hi}$" – Hot-spot-to-top-oil (in tank) gradient at start, "H" - Hot-spot-factor, "g_r" – Average-winding-to-average-oil (in tank) temperature gradient at rated current

Correspondingly, temperature decrease to a level corresponding to a load factor "K" is given by equation (6):

$$\theta_h(t) = \theta_a + \Delta\theta_{or} \times \left[\frac{1 + R \times K^2}{1 + R} \right]^x + \left\{ \Delta\theta_{oi} - \Delta\theta_{or} \times \left[\frac{1 + R \times K^2}{1 + R} \right] \right\} \times f_3(t) + Hg_r K^y$$

Top oil exponent "x" and the winding exponent "y" shall be taken from Table-3.

Where

$f1(t) = \left(1 - e^{(-t)/(k11 \times \tau 0)}\right)$ equation (7), the relative increase of the top-oil temperature rise according to the unit of steady state value.

Where, "k_{11}"-Thermal model consant and "τ_o"-Average oil time constant

$$f_2(t) = k_{21} \times \left(1 - e^{(-t)/(k_{22} \times \tau_w)}\right) - (k_{21} - 1) \times \left(1 - e^{(-t)/\tau_0 / k_{22}}\right)$$

equation (8), the relative increase of the hot-spot-to-hot-oil gradient according to the unit of steady state value

Where, "k_{21}, k_{22}"- Thermal model constant, "τ_w"- Winding time constant

and $f_3(t) = e^{(-t)/(k_{11} \times \tau_0)}$ equation (9), the relative decrease of the top oil-to-ambient gradient according to the unit of the total decrease.

Table - 3 Recommended thermal characteristics for exponential equations

	Distribution transformer	Medium and large power transformer						
	ONAN	ONAN Restricted (see Note)	ONAN	ONAF Restricted (see Note)	ONAF	OF Restricted (see Note)	OF	OD
Oil exponent "x"	0.8	0.8	0.8	0.8	0.8	1.0	1.0	1.0
Winding exponenty "y"	1.6	1.3	1.3	1.3	1.3	1.3	1.3	2.0
Constant "k_{11}"	1.0	0.5	0.5	0.5	0.5	1.0	1.0	1.0
Constant "k_{21}"	1.0	3.0	2.0	3.0	2.0	1.45	1.3	1.0
Constant "k_{22}"	2.0	2.0	2.0	2.0	2.0	1.0	1.0	1.0
Time constant "τo"	180	210	210	150	150	90	90	90
Time constant "τw"	4	10	10	7	7	7	7	7
NOTE: If a winding of an ON or OF-cooled transformer is zigzag-cooled, a radial spacer thickness of less than 3 mm might cause a restricted oil circulation, i.e. a higher maximum value of the function $f2(r)$ than obtained by spacer ≥ 3mm.								

6.2 Differential Equation Solution:

This method makes use of heat transfer differential equations, applicable for arbitrarily time varying load factor "K" and time varying ambient temperature "ϑ_a". They are intended to be the basis of the software to process data in order to define hot-spot temperature as a function of time and consequently the corresponding

insulation life consumption. The differential equations are represented in block diagram form in fig-2, given below.

In figure 2, please note that the inputs are the load factor "K" and the ambient temperature "ϑ_a" on the left. The output is the desired hot-spot temperature "ϑ_h" on the right. The Laplace variable "s" is essentially the derivative operator "d/dt".

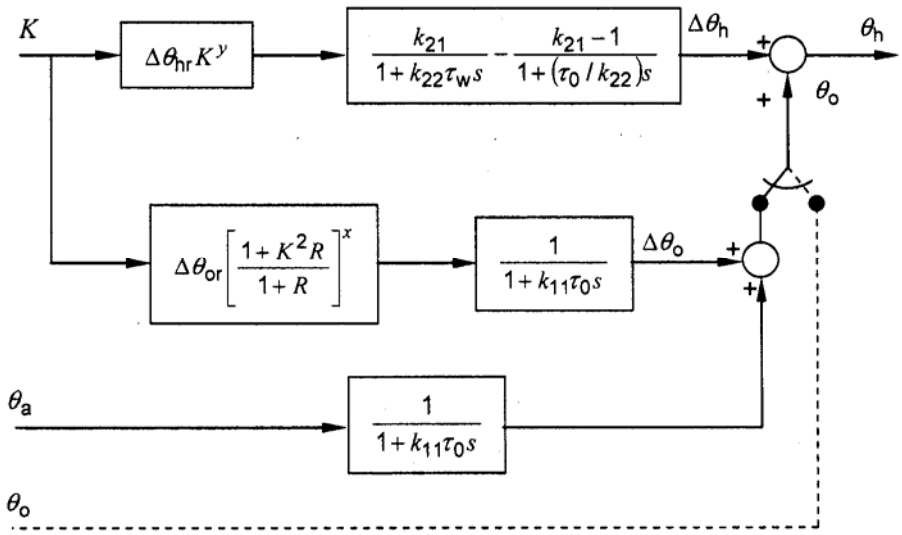


Fig. 2 Block diagram representation of the differential equations.

In Fig.2, the second block in the uppermost path represents the hot-spot dynamics. The first term (with numerator k_{21}) represents the fundamental hot-spot temperature rise, before the effect of changing oil flow past the hot-spot is taken into account. The second term (with numerator $k_{21}-1$) represents the varying rate of oil flow past the hot-spot which changes much more slowly. The combined effect of these two terms is to account for the fact that a sudden rise in load current may cause an otherwise unexpectedly high peak in the hot-spot temperature rise, very soon after the sudden load change. Value of k_{11}, k_{21}, k_{22} and other parameters shall be taken from the Table-3. In the block diagram, "*R*" represents the ratio of supplied load losses at rated current to no-load losses of transformer.

If the top-oil temperature can be measured as an electrical signal into a computing device, then an alternative formulation (shown in figure 2) in dashed line path with the switch in its right position) shall be used instead of the top oil calculation path (switch to the left in figure 2).

The time step shall be less than one half of the smallest time constant "τ_w" to obtain a reasonably accurate value. Further, τ_w and τ_o should not be less than zero.

7.0 PRACTICAL EXAMPLE OF THE EXPONENTIAL EQUATIONS METHOD

7.1 Introduction

The curves in Figure-1 are taken from an example in real life and details of the case will be given in this annex. A 250 MVA, ONAF-cooled transformer was tested as follows. During each time period, the load current was kept constant, that is, the losses changed due to resistance change during each load step. The corresponding flowchart is illustrated in section 7.9.

Table 4 - Load steps of the 250 MVA transformer

Time period min	Load factor
0 - 190	1.0
190 - 365	0.6
365 - 500	1.5
500 - 710	0.3
710 - 735	2.1
735 - 750	0.0

The two main windings were equipped with eight fibre optic sensors each. The hottest spot was found in the innermost main winding (118 kV). In this example the variation of the hottest spot temperature during time period 0 min to 750 min will be defined according to the calculation method described in 6.1. A comparison with the measured curve will be made.

The characteristic data of the transformer, necessary for the calculation, are:

$\theta a = 25{,}6C$

$\Delta\theta or = 38{,}3K$

$R = 1000$ (because the test was made by the short-circuit method)

$H = 1{,}4$ (Hot – spot factor defined by measurement)

$gr = 14{,}5K$

$\tau w = 4{,}6\ to\ 8{,}7\min$ (depending on the loading case. The value in Table 3, that is, 7 min will be used in the calculation)

$\tau o = 162\ to\ 170\min$ (depending on the loading case. The value in Table 3, that is, 150 min will be used in the calculation)

The winding is zigzag-cooled with a spacer separation ≥ 3 mm.

7.2 Time period 0 min to 190 min

$\Delta\theta oi = 12{,}7\,K$

$K = 1{,}0$

$\Delta\theta hi = 0{,}0K$

(This test was started at 08:20 morning. The preceding evening an overloading tet at 1,49p.u had been finished at 22:00)

From Table-3, x= 0.8, y=1.3, k_{11}=0.5, τ_0=150, k_{21}=2.0, τ_w=7

The equations (5), (7) and (8) yield the hot-spot variation as a function of time, hence from equation (5):

$$\theta hi = 25{,}6 + 12{,}7 + \left\{38{,}3 \times \left[\frac{1+1000\times 1{,}0^2}{1+1000}\right]^{0{,}8} - 12{,}7\right\} \times f1(t) + 0{,}0 + \left\{1{,}4 \times 14{,}5 \times 1{,}0^{1{,}3} - 0{,}0\right\} \times f2(t)$$

From equation (7):

$$f1(t) = \left(1 - e^{(-t)/(0.5\times 150)}\right)$$

From equation (8):

$$f2(t) = 2{,}0\left(1 - e^{(-t)/(2{,}0\times 7)}\right) - \left(2{,}0 - 1{,}0\right) \times \left(1 - e^{(-t)/(150/2{,}0)}\right)$$

7.3 Time period 190 min to 365 min

$\Delta\theta oi = 36{,}2\,K$ (calculated in 7.2)

$K = 0{,}6$

$\Delta\theta hi = 22{,}0\,K$ (calculated in 7.2)

The equations (6) and (9) yield the hot-spot variation as a function of time, hence from equation (6):

$$\theta h(t) = 25{,}6 + 38{,}3 \times \left[\frac{1+1000\times 0{,}6^2}{1+1000}\right]^{0{,}8} + \left\{36{,}2 - 38{,}3 \times \left[\frac{1+1000\times 0{,}6^2}{1+1000}\right]^{0{,}8}\right\} \times f3(t) + 1{,}4 \times 14{,}5 \times 0{,}6^{1{,}3}$$

From equation (9):

$$f3(t) = e^{(-t)/(0{,}5\times 150)}$$

7.4 Time period 365 min to 500 min

$\Delta\theta oi = 18{,}84\,K$ (calculated in 7.3)

$K = 1{,}5$

$\Delta\theta hi = 10{,}45\,K$ (calculated in 7.3)

The calculation is identical to that one in 7.2, when the following replacements are made in equation (5):
12,7 replaced by 18,84
1,0 replaced by 1,5
0,0 replaced by 10,45

7.5 Time period 500 min to 710 min

$\Delta\theta oi = 64{,}1\,K$ (calculated in 7.4)

$K = 0{,}3$

$\Delta\theta hi = 37{,}82\,K$ (calculated in 7.4)

The calculation is identical to the one in 7.3, when the following replacements are made in equation (6):

36,2 replaced by 64,1
0,6 replaced by 0,3
22,0 replaced by 37,82

7.6 Time period 710 min to 735 min

$\Delta\theta oi = 9{,}65K$ (calculated in 7.5)

$K = 2{,}1$

$\Delta\theta hi = 4{,}24K$ (calculated in 7.5)

The calculation is identical to the one in 7.4, when the following replacements are made in equation (5):

18,84 replaced by 9,65
1,5 replaced by 2,1
10,45 replaced by 4,24

7.7 Time period 735 min to 750 min

$\Delta\theta oi = 41{,}36K$ (calculated in 7.6)

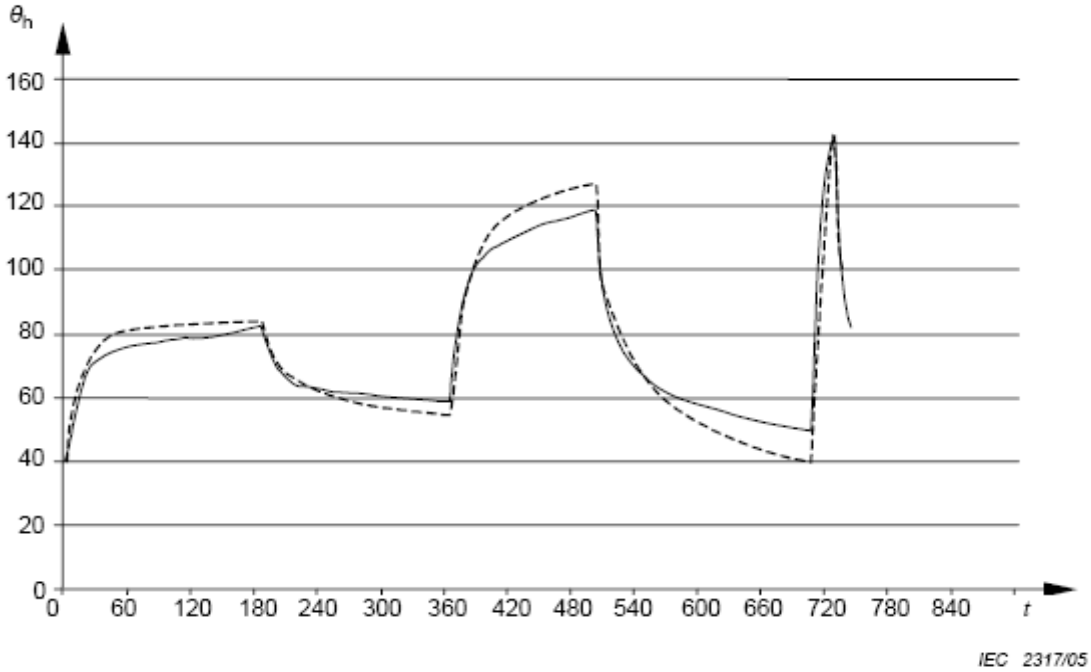
$K = 0{,}0$

$\Delta\theta hi = 71{,}2K$ (calculated in 7.6)

The calculation is made in the same way as in 7.3 and 7.5.

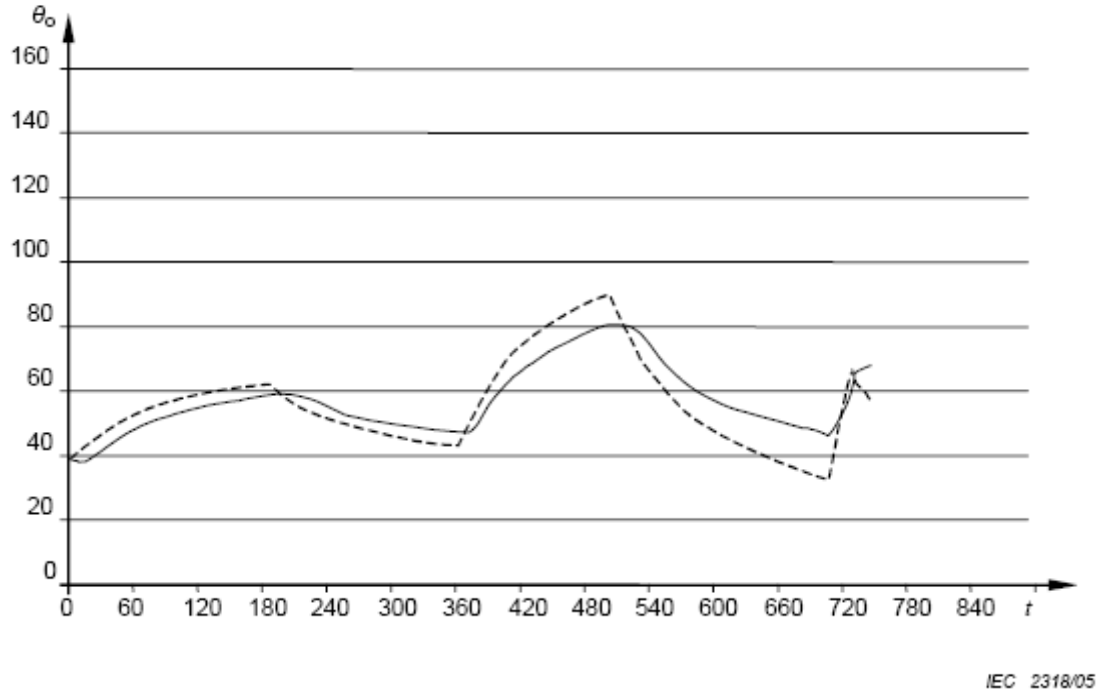
7.8 Comparison with measured values

The calculated and measured hot-spot temperature curves are shown in Figure 3. The corresponding curves for the top-oil temperature are shown in Figure 4. The numerical values at the end of each load step are shown in Table 5.



X-axis time t in minutes Y-axis temperature θ_n in ºC ___ Measured values ----- Calculated values

Fig. 3 Hot-spot temperature response to step changes in the load current



X-axis time t in minutes Y-axis temperature θ_0 in ºC ___ Measured values ----- Calculated values

Fig. 4 Top-oil temperature response to step changes in the load current

Table 5 - Temperatures at the end of each load step

Time (min) / Load factor	Top-oil temperature ºC		Hot-spot temperature ºC	
	Calculated	Measured	Calculated	Measured
190/**1.0**	**61.8**	**58.8**	**83.8**	**82.2**
365/0.6	44.4	47.8	54.9	58.6
500/**1.5**	**89.7**	**80.8**	**127.5**	**119.2**
710/0.3	35.3	46.8	39.5	49.8
735/**2.1**	**67.0**	**65.8**	**138.2**	**140.7**
750/0.0	59.5	68.2	59.5	82.4

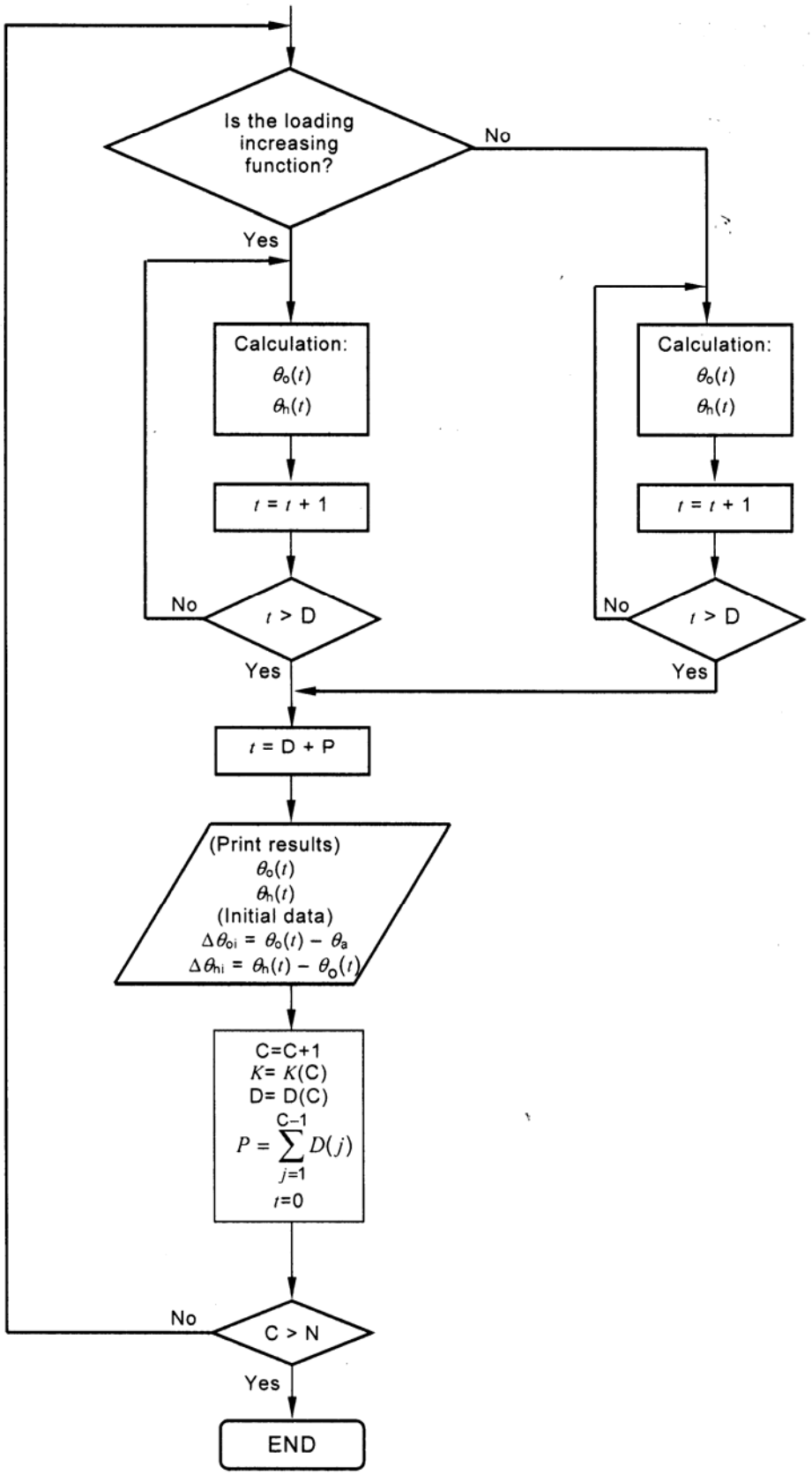
The calculation method is intended to yield relevant values, especially at load increase (noted by bold entries in Table 5).

7.9 Flow chart of example illustrated in section 7.8 is as follows:

START

INPUT DATA
Cooling method: ONAF
The ambient temperature: θ_a = 25,6 °C
The top-oil temperature rise at the rated current: $\Delta\theta_{or}$ = 38,3 K
Ratio of load losses at rated current to no-load losses: R = 1 000
The hot-spot factor: H = 1,4
The winding-to-oil temperature difference: g_r = 14,5 K
The winding time constant: τ_w = 7 min
The average oil time constant: τ_o= 150 min
The oil temperature exponent: x = 0,8
The winding temperature exponent: y = 1,3
Thermal constants: k_{11} = 0,5; k_{21} = 2,0; k_{22} = 2,0
The number of intervals in load cycle: N = 6
Load factor for each interval:
K(1) = 1,0; I(1) = (0 – 190) min
K(2) = 0,6; I(2) = (190 – 365) min
K(3) = 1,5; I(3) = (365 – 500) min
K(4) = 0,3; I(4) = (500 – 710) min
K(5) = 2,1; I(5) = (710 – 735) min
K(6) = 0,0; I(6) = (735 – 750) min
Duration of each interval:
D(1) = 190 min
D(2) = 175 min
D(3) = 135 min
D(4) = 210 min
D(5) = 25 min
D(6) = 15 min
Spacer thickness ≥ 3 mm

C = 1 (interval counter settings)
K = K(1) D = D(1)
P = 0 t =0
(Initial data)
$\Delta\theta_{oi}$ = 12,7 K $\Delta\theta_{hi}$ = 0,0 K


Is the loading increasing function?
No
Yes
Calculation:
$\theta_o(t)$
$\theta_h(t)$
$t = t + 1$
No
$t > D$
Yes
Calculation:
$\theta_o(t)$
$\theta_h(t)$
$t = t + 1$
No
$t > D$
Yes
$t = D + P$
(Print results)
$\theta_o(t)$
$\theta_h(t)$
(Initial data)
$\Delta\theta_{oi} = \theta_o(t) - \theta_a$
$\Delta\theta_{hi} = \theta_h(t) - \theta_o(t)$
C=C+1
$K = K(C)$
D= D(C)
$P = \sum_{j=1}^{C-1} D(j)$
$t=0$
No
C > N
Yes
END

8.0 BIBILIOGRAPHY

8.1 IEC 60076-7- Loading Guide for Oil immersed Power Transformers

8.2 Permissible overloading of Oil immersed transformers and Regulators, United States department of interior bureau of reclamation